

JEE Mains 2019 Chapter wise Question Bank

Magnetism - Questions

Q1

A bar magnet is demagnetized by inserting it inside a solenoid of length 0.2 m, 100 turns, and carrying a current of 5.2 A. The coercivity of the bar magnet is:

- (1) 285 A/m (2) 2600 A/m
(3) 520 A/m (4) 1200 A/m

9 Jan Morning

Q2

A solid metal cube of edge length 2 cm is moving in a positive y -direction at a constant speed of 6 m/s. There is a uniform magnetic field of 0.1 T in the positive z -direction. The potential difference between the two faces of the cube perpendicular to the x -axis, is:

- (1) 12 mV (2) 6 mV
(3) 1 mV (4) 2 mV

10 Jan Morning

Q3

A magnet of total magnetic moment $10^{-2} \hat{i}$ A-m² is placed in a time varying magnetic field, $B \hat{i} (\cos \omega t)$ where $B = 1$ Tesla and $\omega = 0.125$ rad/s. The work done for reversing the direction of the magnetic moment at $t = 1$ second, is:

- (1) 0.01 J (2) 0.007 J
(3) 0.028 J (4) 0.014 J

10 Jan Morning

Q4

An insulating thin rod of length l has a linear charge density $\rho(x) = \rho_0 \frac{x}{l}$ on it. The rod is rotated about an axis passing through the origin ($x = 0$) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is:

- (1) $\pi n \rho l^3$ (2) $\frac{\pi}{3} n \rho l^3$
(3) $\frac{\pi}{4} n \rho l^3$ (4) $n \rho l^3$

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Q5

A hoop and a solid cylinder of same mass and radius are made of a permanent magnetic material with their magnetic moment parallel to their respective axes. But the magnetic moment of hoop is twice of solid cylinder. They are placed in a uniform magnetic field in such a manner that their magnetic moments make a small angle with the field. If the oscillation periods of hoop and cylinder are T_h and T_c respectively, then:

- (1) $T_h = T_c$ (2) $T_h = 2 T_c$
(3) $T_h = 1.5 T_c$ (4) $T_h = 0.5 T_c$

10 Jan Evening

Q6

A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of 20×10^{-6} J/T when a magnetic intensity of 60×10^3 A/m is applied. Its magnetic susceptibility is:

- (1) 3.3×10^{-2} (2) 4.3×10^{-2}
(3) 2.3×10^{-2} (4) 3.3×10^{-4}

11 Jan Evening

Q7

A paramagnetic material has 10^{28} atoms/m³. Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4} . Its susceptibility at 300 K is:

- (1) 3.267×10^{-4} (2) 3.672×10^{-4}
(3) 3.726×10^{-4} (4) 2.672×10^{-4}

12 Jan Evening

Q8

The mean intensity of radiation on the surface of the Sun is about 10^8 W/m². The rms value of the corresponding magnetic field is closest to :

- (1) 1 T (2) 10^2 T
(3) 10^{-2} T (4) 10^{-4} T

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Q9

A passenger train of length 60 m travels at a speed of 80 km/hr. Another freight train of length 120 m travels at a speed of 30 km/h. The ratio of times taken by the passenger train to completely cross the freight train when: (i) they are moving in same direction, and (ii) in the opposite directions is:

- (1) $\frac{11}{5}$ (2) $\frac{5}{2}$
(3) $\frac{3}{2}$ (4) $\frac{25}{11}$

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Magnetism - Answers

Q1

(2) Corecivity, $H = \frac{B}{\mu_0}$ and $B = \mu_0 n i \left(n = \frac{N}{\ell} \right)$

or, $H = \frac{N}{\ell} i = \frac{100}{0.2} \times 5.2 = 2600 \text{ A/m}$

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Q2

(1) Potential difference between two faces perpendicular to x-axis

$$= \lambda V.B = 2 \times 10^{-2} (6 \times 0.1) = 12 \text{ mV}$$

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Q3

(Bonus)

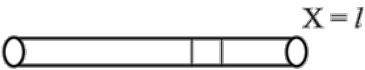
Work done, $W = 2 \text{ mB}$

$$= 2 \times 10^{-2} \times 1 \cos(0.125)$$

$$= 0.02 \text{ J}$$

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Q4

(3) 

$1 \leftarrow x \rightarrow dx$

Magnetic moment, $M = NIA$

$$dQ = \rho dx$$

$$dI = \frac{dQ}{2\pi} \cdot \omega$$

$$dM = dI \times A$$

$$= \frac{\omega}{2\pi} \cdot \frac{\rho_0}{\ell} \cdot x \pi x^2 dx$$

$$\Rightarrow M = \frac{\rho_0}{\ell} n \pi \int_0^{\ell} x^3 dx$$

$$= \frac{\pi}{4} \cdot n \rho \ell^3$$

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Q5

(1) Using, time / oscillation period,

$$T = 2\pi \sqrt{\frac{I}{MB}}$$

Where, M = magnetic moment, I moment of inertia and B = magnetic field

$$T_h = 2\pi \sqrt{\frac{mR^2}{(2MB)}}$$

$$T_c = 2\pi \sqrt{\frac{1/2mR^2}{MB}}$$

Clearly, $T_h = T_c$

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Magnetism

Q6

(4) Magnetic susceptibility,

$$\chi = \frac{I}{H}$$

$$\text{where, } I = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{20 \times 10^{-6}}{10^{-6}}$$

$$= 20 \text{ N/m}^2$$

$$\text{Now, } \chi = \frac{20}{60 \times 10^3} = \frac{1}{3} \times 10^{-3} = 3.3 \times 10^{-4}$$

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Q7

(1) According to Curie law for paramagnetic substance,

$$\chi \propto \frac{1}{T_C}$$

$$\frac{\chi_1}{\chi_2} = \frac{T_{C2}}{T_{C1}}$$

$$\frac{2.8 \times 10^{-4}}{\chi_2} = \frac{300}{350}$$

$$\chi_2 = \frac{2.8 \times 350 \times 10^{-4}}{300}$$

$$= 3.266 \times 10^{-4}$$

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Q8

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$$(4) I = \frac{B_0^2}{2\mu_0} \cdot C$$

$$\Rightarrow \frac{B_0^2}{2} = \frac{I\mu_0}{C}$$

$$\Rightarrow B_{\text{rms}} = \sqrt{\frac{I\mu_0}{C}}$$

$$= \sqrt{\frac{10^8 \times 4\pi \times 10^{-7}}{3 \times 10^8}}$$

$$\approx 6 \times 10^{-4} \text{ T}$$

Which is closest to 10^{-4} .

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Q9

(Bonus) We have, $T = 2\pi \sqrt{\frac{I}{MB_x}}$

$$\therefore \frac{T_1^2}{T_2^2} = \frac{Bx_2}{Bx_1}$$

$$\text{or } \left(\frac{2}{1.5}\right)^2 = \frac{B_2 \cos 45^\circ}{B_1 \cos 30^\circ} = \frac{B_2 \times 2}{\sqrt{2} \times B_1 \times \sqrt{3}}$$

$$\left(\frac{4}{3}\right)^2 = \frac{B_2}{B_1} \times \frac{2}{\sqrt{6}}$$

$$\therefore \frac{B_1}{B_2} = \frac{9}{8\sqrt{6}} = 0.46$$

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